MD KASHIF ALAM

Robotics Software Engineer | ROS2, Autonomous Navigation & Control

Email | Phone | Portfolio | LinkedIn | GitHub | YouTube

ABOUT ME

Robotics Software Engineer focused on autonomous navigation, motion planning, control, and real-time robotic systems. Hands-on experience with ROS2, C++, Python, Nav2, SLAM, trajectory tracking, sensor fusion, state estimation, and robot simulation. Experienced in integrating LiDAR/IMU/GPS and embedded systems, developing navigation and control algorithms, and debugging hardware-software systems. Built autonomous mobile robots, quadruped robotics, UAV flight-control systems, and ROS2 navigation stacks.

EDUCATION

Bachelor of Science in AI & ML IIT Patna / 2027

SKILLS

- **Robotics & autonomy:** ROS2, Nav2, SLAM, Localization, Sensor Fusion, Autonomous Navigation, Path Planning, Motion Planning, Trajectory Tracking, Obstacle Avoidance, TF2
- **Programming:** C++, Python, C, Bash
- **Robotics Control & Estimation:** PID Control, Kinematics, Dynamics, State Estimation, Attitude Estimation, Sensor Calibration, Coordinate Transformations
- **Perception & Sensors:** LiDAR, IMU, GPS/GNSS, Magnetometer, Barometer, Encoders, Camera Sensors, Sensor Fusion
- **Simulation & Robotics Tools:** Gazebo, RViz, MoveIt2, URDF/Xacro, MATLAB/Simulink, OpenCV
- **Embedded & Hardware:** STM32, ESP32, Arduino, Raspberry Pi, RTOS, UART, I2C, SPI, CAN, PWM, ESCs, BLDC Motors
- **Development:** Linux/Ubuntu, Git/GitHub, Docker, VS Code, CMake, WSL

EXPERIENCE

Technical Intern – India Space Lab (ISL) Jan 2026 – March 2026
AI & ML Engineer Intern – Springer Capital Feb 2026 – May 2026

PROJECTS

- **ROS2 Nav2 Autonomous Navigation Stack** *Project Link*
  - Implemented an end-to-end autonomous navigation system using ROS2 Nav2 with AMCL localization, global planning, and DWB-based local control. Integrated sensor and odometry data to enable autonomous navigation and obstacle-aware motion in a simulated robotic environment**.**
- **Industrial Robot / FANUC** *Project Link*
  - Configured a FANUC CRX-10iA with ROS2/MoveIt2 for motion planning, trajectory execution, and industrial robot control.
- **Custom DWA Local Planner (ROS2)** *Project Link*
  - Developed a custom Dynamic Window Approach local planner in ROS2 for obstacle-aware motion planning using LiDAR and odometry. Implemented velocity-space trajectory evaluation and cost-based selection to generate collision-free local motion commands.
- **Trajectory Tracking Controller** *Project Link*
  - Developed a ROS2-based trajectory tracking controller to follow predefined waypoints using real-time odometry and velocity feedback.
- **Custom UAV Flight Controller & Avionics Firmware** *Project Link*
  - Developed a custom UAV flight-controller firmware integrating IMU, magnetometer, GPS, and barometer sensors for real-time attitude estimation, navigation, and flight control. Implemented sensor calibration, sensor fusion, PID-based control, actuator/ESC control, telemetry, and real-time control-loop processing on embedded hardware. Designed the firmware architecture to process sensor data, estimate vehicle state, generate control outputs, and drive the propulsion system in real time.
- **Multi-Agent 3D Path Planning & Autonomous UAV Mission Planner** *Project Link*
  - Developed a multi-agent UAV mission planner implementing 3D path planning, collision avoidance, and autonomous waypoint navigation. Designed planning and coordination algorithms for multiple UAVs operating in shared environments, with support for mission generation and autonomous flight execution.
- **Sensor Fusion & UAV Attitude Estimation** *Project Link*
  - Developed a real-time attitude-estimation pipeline using gyroscope, accelerometer, and magnetometer measurements. Implemented sensor calibration, coordinate transformations, roll/pitch estimation, tilt-compensated heading, and complementary sensor fusion while accounting for sampling time and sensor-axis conventions